DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (A)
Osmania University, Hyderabad- 07

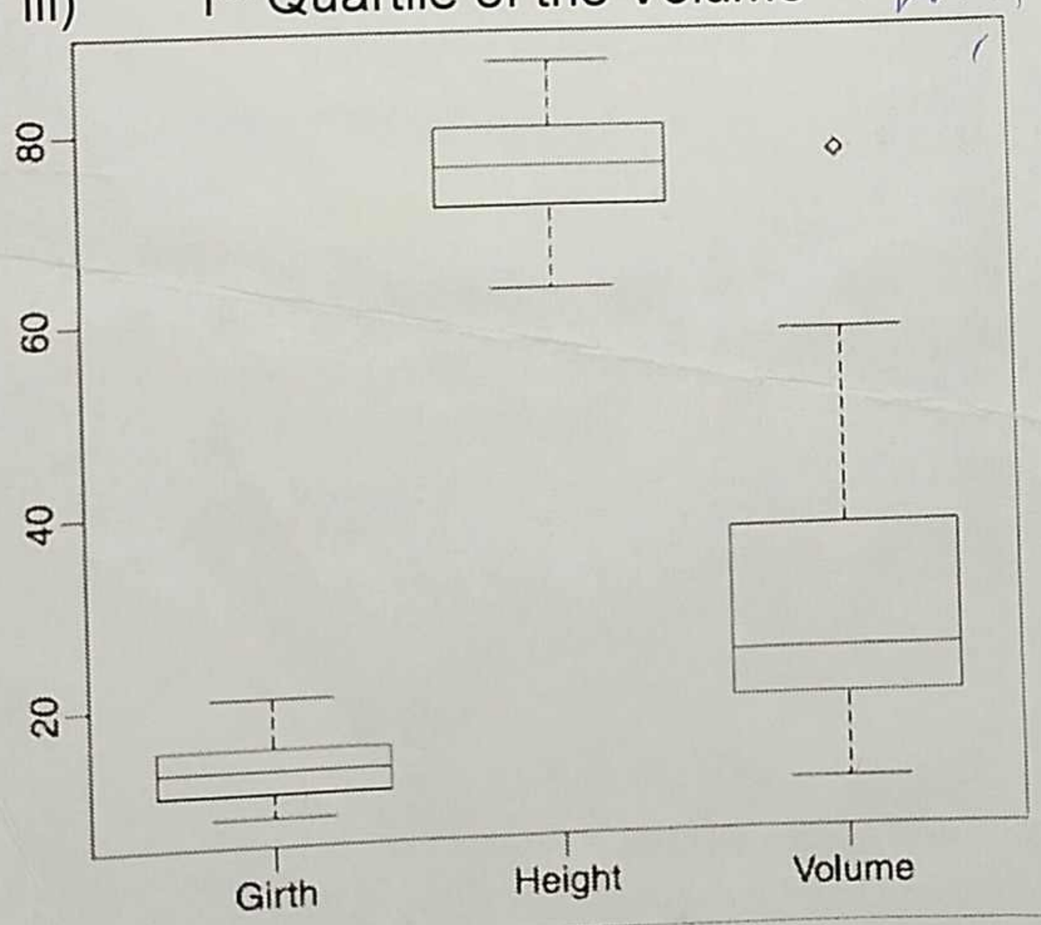
CLASS TEST-II

Subject : Data Analytics using R

Class: B.E. (CSE)
Semester: IV
Academic Year: 2024-2025

Max Marks: 30
Duration:1.5 hr(2:30 p.m. to 4:00 p.m.)
Date : 27-06-2025

Note: If any data missing may be assumed
Answer all questions in Part A

Q No	PART A Answer All Questions (5 x 2 = 10 Marks)	Marks	BT	CO
1.	What is clustering? List out the algorithms used in Clustering?	2	1	1
2.	Calculate the **Confidence** of the following association rule for the given following purchase data **{Bread, Butter} => {Milk}** **TID** / **Items** 1 — Bread, Butter, Milk 2 — Bread, Butter 3 — Beer, Cookies, Diapers 4 — Milk, Diapers, Bread, Butter	2	5	4
3.	What is the purpose of the following functions in R and how these can be used in parallel computing with R i. foreach() ii. mclapply()	2	3	3
4.	Please identify what is the following graph and identify the approx. values for the following statistical measures based on the visual diagram i) Maximum of Girth ii) Mean of the Height iii) 1st Quartile of the Volume 	2	4	1

5.	Following is the R program for Time Series data. What will be the output after running the following program. Please write down the output: install.packages("forecast") install.packages("tseries") library(forecast) library(tseries) sales_data <- c(200, 220, 210, 230, 250, 270, 300, 320, 310, 330, 350, 370, 210, 230, 220, 240, 260, 280, 310, 330, 320, 340, 360, 380) sales_ts <- ts(sales_data, start = c(2023, 1), frequency = 12) print(sales_ts)	2	6	
	PART B Answer Any Two Questions (2 x 10 = 20Marks)			
	a) What is K means algorithm? What is its application	3	2	4
6.	b) We have 4 medicines as our training data points object and each medicine has 2 attributes. Each attribute represents coordinate of the object. The object being to know which medicines belong to which cluster. Calculate the distance matrix for **2 iterations** only for the following given data. **Object** | Attribute1 (X): weight index | Attribute2 (Y): pH Medicine A | 1 | 1 Medicine B | 2 | 1 Medicine C | 4 | 3 Medicine D | 5 | 4	7	5	4

→ 9989253132

	example	gain (Infogain)? Explain	3	2	3
7.	b) For the following data: i)Calculate the entropy of the total dataset ii)Calculate the Infogain for splitting the dataset on 'Gender' attribute"		7	6	3
	a) What is Text Mining and explain the text mining process? Discuss the following Text mining methodologies i) Sentiment Analysis ii) Entity Recognition		6	3	3

8	b) Following is the R program for Parallel computing. Please list out what will be the output of the following:
	
	Load the parallel package
	library(parallel)
	# Define sample data: a list of numbers 1 to 5
	numbers <- list(1, 2, 3, 4, 5)
	# Define a function to square each number
	
	square <- function(x) {
	Sys.sleep(1) # simulate some delay
	return(x^2)
	}
	
	# Apply the function in parallel using mclapply
	result <- mclapply(numbers, square, mc.cores = 2)
	# Print the result
	print(result)
			4	6	3

Sl No	Gender	Car Ownership	Travel Cost	Income Level	Transportation
1	Male	0	Cheap	Low	Bus
2	Male	1	Cheap	Medium	Bus
3	Female	1	Cheap	Medium	Train
4	Female	0	Cheap	Low	Bus
5	Male	1	Cheap	Medium	Bus
6	Male	0	Standard	Medium	Train
7	Female	1	Standard	Medium	Train
8	Female	1	Expensive	High	Car
9	Male	2	Expensive	Medium	Car
10	Female	2	Expensive	High	Car